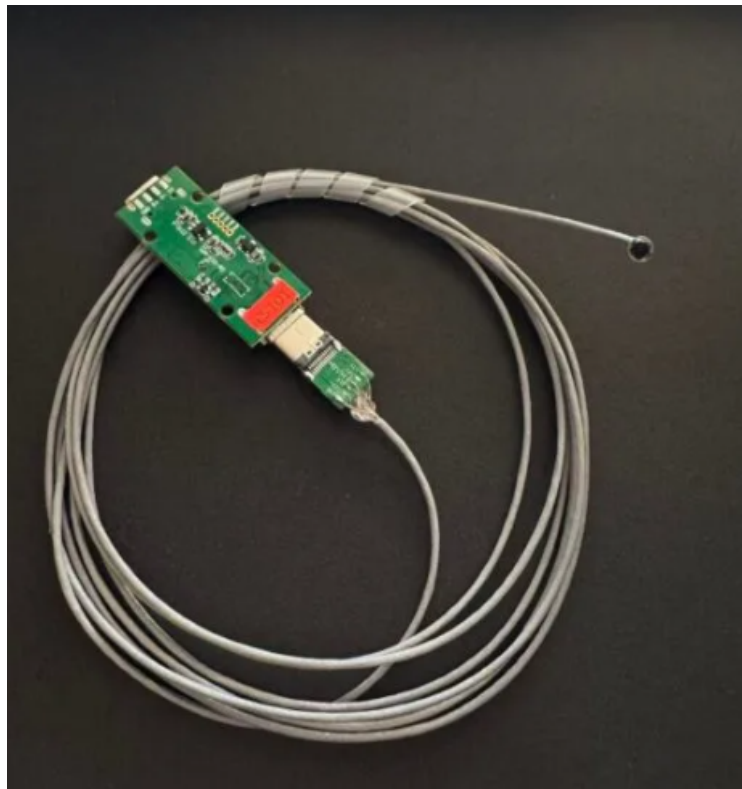


VisionMeta

VM-YJES101X Specification



Content

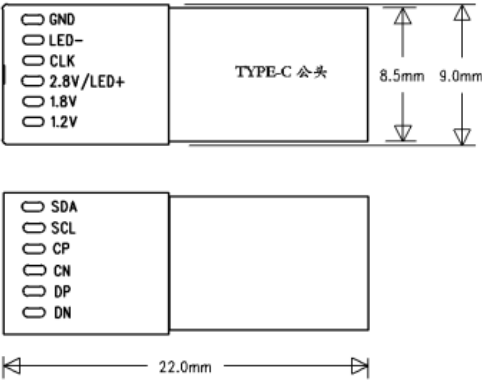
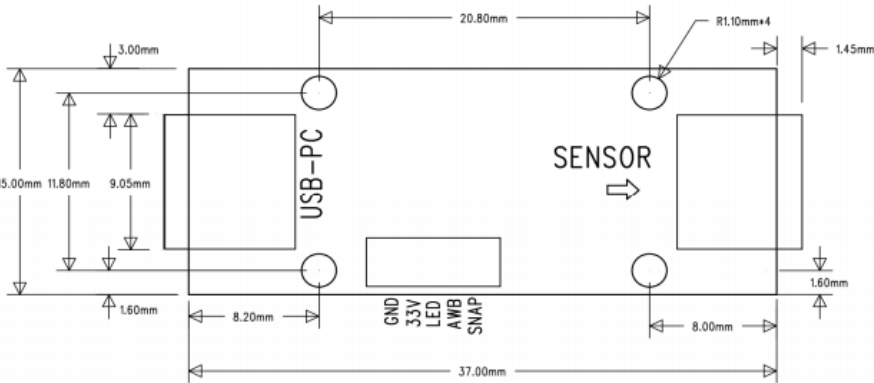
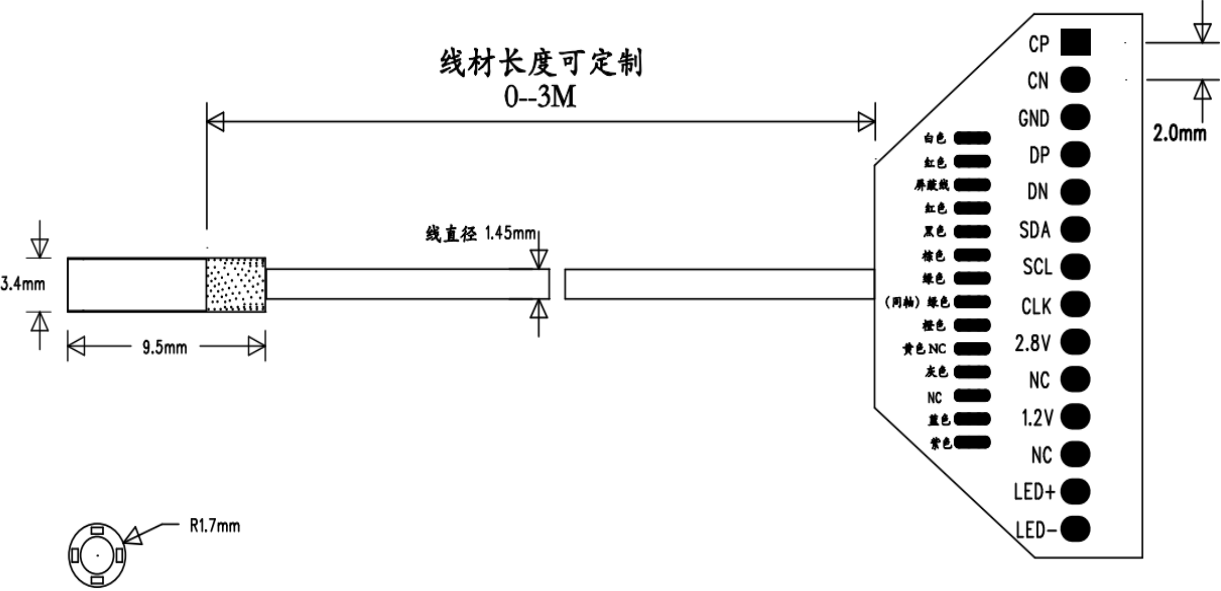
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Module Specification

NO	Item	Specification
1	Module No.	VM-YJES101X
2	Image Sensor	ES101
3	Diameter	3.0mm
4	Lens Size	1/9inch
5	Pixel Size	1.4um X 1.4um
6	Resolution	1000 H×1000 V
7	Sensitivity	0.92V/Lux-sec
8	S/N Radio	37.2 dB
9	Dynamic Range	72db
10	LED	4 pcs LED Light
11	Support Resolution and Frame Rate	25fps Mjpeg@72Mhz 15fps VGA YUV@72Mhz
12	Support Drive - Free Protocols	USB Video Class (UVC)
13	Interface	USB2.0 HS/FS
14	Output Format	MJPEG / YUV2 (YUYV)
15	Auto Exposure Control	Support
16	Adjustable Parameter	Brightness, Contrast Ratio, Color Saturation, Tone, White Balance, Exposure Degree
17	Operating Temperature	-30℃ ~ 70℃
18	Stable Temperature	0℃ ~ 50℃

Module Diagram

CP	双绞白色	双绞
CN	双绞红色	双绞
GND	屏蔽线同轴屏蔽线	
DP	双绞红色	双绞
DN	双绞黑色	双绞
SDA	棕色	
SCL	绿色	
CLK	(同轴) 绿色	
2.8V	橙色	
1.2V	灰色	
LED+	蓝色	
LED-	紫色	
SENSOR型号: ES101		
备注: 灰色线 直径1.45MM		
分辨率: 1000X1000 视角 140°		
景深: 3-50mm		
线材长度可选: 0-3米		
LED: PB0301WE-010F030		
LED 色温: 4500-5000K		
4颗LED灯并联 电压: 2.6V-3V		
钢管直径3.4MM 长度9.5MM		
防水等级: 头端防水 IPX7		
储存温度: 0℃~50℃		
头端工作温度: -10℃~+70℃		



Functional Testing & Standards

NO.	Test Items	Standard	Testing Method	Testi ng Resu lt
1	Dust	There is no dirt on the surface of the module, no scratches visible to the naked eye, and no impurities in the plastic.	Small light box, 6500K white backlight	A
2	Imaging Direction	Meet Drawing Requirements	Digimatic Calipers	A
3	Dead&Wound Pixel	1)Under dark field: the total number of bright spots is less than 400pixels; the clustered bright spots of 2 adjacent pixels are less than 20; there are no clustered bright spots of more than 3 pixels. 2) Under the white field: the total number of dark spots is less than 400pixels; the number of clustered dark spots of 2 adjacent pixels is less than 20; there are no clustered dark spots of more than 3 pixels.	Small light box, 6500K white backlight	A
4	Resolution	1 million pixels: the center is not less than 400LW/PH	MTF test procedure	A
5	FOV	Diagonal:90°±3°	Digimatic Calipers、Fov testing board	A
6	Shading	The average brightness of the four corners is 90 % away from the center of the test site, at least reaching 45% of the brightness of the center test site; the brightness of each corner reaches 45% of the center brightness	Testing software、light box, (3400Lux, 6500K white backlight	A
7	Color	The color reproduction error is not more than 20%	Testing software Colorchecker card	A
8	TV Distortion	Neither pincushion (positive numbers) nor barrel distortion (negative numbers) is significant at 1%	Testing software、ISO12233testing board	A
9	Grayscale	The brightness difference between two adjacent gray scales that can be distinguished must be less than 8	Testing software、Grayscale testing board	A
10	Chromatic Aberration	Less than 0.5	Testing software、ISO12233testing board	A

Reliability Test Standards

Item	Testing Condition
1.Drop Test	Height: 1500mm, six sides;
2.Vibration Test	30Hz, 3g, 3axis, 60min/axis;
3.Storage Test	High and low temperature: -20~+70
5.Working environment/ humidity	-10℃→50℃ /95%
6.High and low temperature test	-10℃/4 hours70℃/4 hours

Note: After the product has passed various reliability tests, the size, appearance, torque, and optical properties all meet the product requirements.

Description of non-conforming product classification:

- a) Seriously unqualified: unqualified batches determined by inspection, or unqualified products that cause large economic losses and directly affect product quality, main functions, performance technical indicators, etc., are represented by the letter A;
- b) General unqualified: Generally unqualified refers to unqualified products that can meet the following conditions at the same time, and is represented by the letter B.

Packing Specifications

1. After checking the product is qualified, put it into the blister tray (100PCS/pan);
2. Pack every 5 trays into a vacuum packaging bag, and put it into a bag of moisture-proof beads, and seal it with a packaging machine;
3. Pack every two packs into a custom-made carton, seal it with transparent tape, and fill in the label.

